

Institute of Actuaries of India

Subject CT8 – Financial Economics

October 2014 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i. Pay off Diagram for European Call:**

		51.8	u	1.06
	20.000	561.8	d	0.95
	530.000		r	0.05
500		503.5	t	3 months = 0.25
	475			$[\exp(r \cdot T) - d] / (u - d) =$
	0	451.25	p	0.5689
		0	1-p	0.4311

Value of the 6 month European Call = $\exp(-0.5 \cdot 0.05) \cdot 51.8 \cdot 0.5689^2 = 16.351$

[3]**ii. The pay off diagram for European Put:**

	B	0
	0	561.8
	530.000	
500		503. 56.50
	475	
	35	451.25
	C	58.75

Value of European put option = $\exp(-0.05 \cdot .5) \cdot [0.4311^2 \cdot 58.75 + 2 \cdot 0.4311 \cdot 0.5689 \cdot 6.50] = 13.759$

Using Put call parity, Value of put + stock = $500 + 13.759 = 513.759$

Value of call + discounted price = $16.351 + 510 \cdot \exp(-0.05 \cdot .5) = 513.759$

[3]**iii. For the American option, the value at final nodes is same for a European Option. At earlier nodes, the value of the option is greater of value calculated at the node and the payoff from early exercise.**

We can find that the value at node B = $6.50 \cdot \exp(-0.25 \cdot 0.05) \cdot 0.4311 = 2.767$.

Value at node C = $\exp(-0.25 \cdot 0.05) \cdot (6.50 \cdot 0.5689 + 58.75 \cdot 0.4311) = 28.65$

However, payoff from immediate exercise is 35 at node C. Hence it is advisable to exercise the put option at this point.

Value of the American put option at time $t=0$ will be

$\exp(-0.05 \cdot .25) \cdot (35 \cdot 0.4311 + 2.767 \cdot 0.5689) = 16.46$

[3]

- iv. Expected payoff in 3 months time is calculated by using real rate of return of 9%.
 $p=0.6614$. Hence expected payoff = $20 \times 0.6614 = 13.228$. Unfortunately, it is not easy to know the correct discount rate to apply to the expected payoff in real world to be able to compute the value of the option. Risk neutral valuation solves this problem as under risk neutral valuation all assets are expected to earn the risk free rate.

[2]

[11 Marks]

Solution 2:

- i. Let $f = f(S_t, t) = S^k$

$$\frac{\partial f}{\partial S} = kS^{k-1}, \quad \frac{\partial^2 f}{\partial S^2} = k(k-1)S^{k-2}, \quad \frac{\partial f}{\partial t} = 0$$

$$\begin{aligned} \text{Using Ito calculus, } df &= kS^{k-1}dS_t + \frac{1}{2}k(k-1)S^{k-2}(dS_t)^2 \\ &= f[k\mu + \frac{1}{2}k(k-1)\sigma^2]dt + f[k\sigma]dW_t \end{aligned}$$

Hence $f = S^k$ follows Geometric Brownian motion, with drift $\mu' = k\mu + \frac{1}{2}k(k-1)\sigma^2$ and volatility $\sigma' = k\sigma$.

$$\text{Hence } f_t = f_0 e^{(\mu' - \frac{1}{2}\sigma'^2)t + \sigma'W_t}$$

$$\text{This means } \frac{f}{f_0} \sim \text{lognormal}((\mu' - \frac{1}{2}\sigma'^2)t, \sigma'^2 t)$$

The mean and variance of a log normal distribution with parameters μ and σ^2

$$\text{Is given by } e^{\mu + \frac{1}{2}\sigma^2} \text{ and } (e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$$

$$\text{This means } E(f) = f_0 e^{\mu' t} \text{ and } V(f) = f_0^2 e^{2\mu' t} (e^{\sigma'^2 t} - 1)$$

[5]

- ii. Let $f = f(t, S_t) = e^{-rt}S_t$

$$\frac{\partial f}{\partial t} = -re^{-rt}S_t, \quad \frac{\partial f}{\partial S} = e^{-rt}, \quad \frac{\partial^2 f}{\partial S^2} = 0$$

$$\begin{aligned} \text{By Ito calculus, } df &= -re^{-rt}S_t dt + e^{-rt}dS_t \\ &= (\mu - r)f dt + \sigma f dW_t \quad (\text{on substituting the expression for } dS_t) \end{aligned}$$

which is a martingale if and only if $\mu = r$

[3]

- iii. Combining results from part (a) and (b), we need, for discounted S^k to be a martingale,
- $$k\mu + \frac{1}{2}k(k-1)\sigma^2 = r$$

For given values of r , σ and k , one can solve for the value of μ for which discounted S^k will be a martingale.

[1]

[9 Marks]

Solution 3:

i.

	Mean	SD	Split
Growth Stock	17%	25%	0.7
Value Stock	13%	12%	0.3
Risk free rate	6%		
Correlation	0.5		

Return on market portfolio

$$R_M = 0.3 \times 13\% + 0.7 \times 17\% = 15.8\%$$

$$\sigma_M = \sqrt{(0.3^2 \times 0.12^2 + 0.7^2 \times 0.25^2 + 2 \times 0.3 \times 0.7 \times 0.25 \times 0.12 \times 0.5)} = 19.55\%$$

[2]

ii.

$$\beta_{Growth} = \frac{Cov(r_{Growth}, r_m)}{\sigma_m^2} = \frac{Cov(r_G, 0.70r_G + 0.30r_V)}{\sigma_m^2}$$

$$\begin{aligned} \beta_{Growth} &= \frac{0.7\sigma_G^2 + 0.30 \times Cor(r_G, r_V) \times \sigma_G \times \sigma_V}{\sigma_m^2} = \frac{0.7 \times 25^2 + 0.30 \times 0.5 \times 25 \times 12}{19.55^2} \\ &= \frac{482.50}{19.55^2} = 1.26 \end{aligned}$$

$$\beta_{Value} = \frac{Cov(r_{Value}, r_m)}{\sigma_m^2} = \frac{Cov(r_V, 0.70r_G + 0.30r_V)}{\sigma_m^2}$$

$$\begin{aligned} \beta_{Value} &= \frac{0.3\sigma_V^2 + 0.70 \times Cor(r_G, r_V) \times \sigma_G \times \sigma_V}{\sigma_m^2} = \frac{0.3 \times 12^2 + 0.70 \times 0.5 \times 25 \times 12}{19.55^2} \\ &= \frac{148.20}{19.55^2} = 0.39 \end{aligned}$$

[2]

iii. Residual Variance of Growth Stock = $25^2 - 1.26^2 \times 19.55^2 = 15.89\%^2$
 Residual Variance of Growth Stock = $12^2 - 0.39^2 \times 19.55^2 = 86.54\%^2$

[2]

iv. $E(r_{Growth}^{CAPM}) = 6 + 1.26 \times (15.80 - 6) = 18.37\% \neq 17\%$

$$E(r_{Value}^{CAPM}) = 6 + 0.39 \times (15.80 - 6) = 9.80\% \neq 13\%$$

CAPM does not hold in this economy.

[2]

[8 Marks]

Solution 4:

- i. Let $X(t)$ be a continuous time stochastic process with respect to filtration F_t where $0 \leq t < \infty$. The process is a martingale if $E[X(t)] < \infty$ for all t and $E[X(t) | F_s] = X(s)$ for all $s < t$.

[1]

ii. $f(0, t) = \int_0^t f(s) dB_s \sim N(0, \int_0^t f^2(s) ds)$

Hence $\int_u^t f(s) dB_s = f(0, t) - f(0, u)$

Since both $f(0, t)$ and $f(0, u)$ are Normal variables, their linear combination also follows Normal distribution.

$$E(f(u, t)) = E(f(0, t)) - E(f(0, u)) = 0$$

$$\text{Var}(f(u, t)) = \text{Var}(f(0, t)) + \text{Var}(f(0, u)) - 2\text{cov}(f(0, t), f(0, u))$$

$$\text{Now } \text{cov}(f(0, t), f(0, u)) = \text{cov}(f(0, u) + f(u, t), f(0, u))$$

$$= \text{cov}(f(0, u), f(u, t)) + \text{Var}(f(0, u))$$

$$= \text{Var}(f(0, u)), \text{ since } f(0, u) \text{ and } f(u, t) \text{ are independent}$$

$$\text{Hence } \text{Var}(f(u, t)) = \text{Var}(f(0, t)) - \text{Var}(f(0, u))$$

[2]

iii. $E(M_{0,t} | M_{0,s}, t > s) = E(M_{0,s} M_{s,t} | M_{0,s}, t > s) = E \left[M_{0,s} e^{\int_s^t f(s) dB_s - \frac{1}{2} \int_s^t f^2(s) ds} \middle| M_{0,s} \right]$

$$= M_{0,s} e^{-\frac{1}{2} \int_s^t f^2(s) ds} E \left[e^{\int_s^t f(s) dB_s} \right]$$

$$= M_{0,s} e^{-\frac{1}{2} \int_s^t f^2(s) ds} e^{\frac{1}{2} \int_s^t f^2(s) ds} = M_{0,s}$$

Using Moment Generating Function of Normal, and the result in (b)

[3]

[6Marks]

Solution 5:

For a non-dividend paying stock, put call parity gives,

$$P + S = C + Ke^{-r(T-t)}$$

Where P = value of European put option, S = stock price underlying the call option, C = value of European call option, K = strike price.

- i. Delta is the rate of change of option price with respect to price of the underlying asset/security. Differentiating the above equation with respect to S:

$$\frac{\partial P}{\partial S} + 1 = \frac{\partial C}{\partial S}$$

Or $\frac{\partial P}{\partial S} = \frac{\partial C}{\partial S} - 1$ which shows delta of European put equals the delta of the corresponding European call option less 1.

[1]

- ii. The gamma of a portfolio on an underlying stock is the rate of change of portfolio's delta with respect to the price of the underlying security/asset. Differentiating with respect to S again, we get gamma of European call option equals gamma of European put.

$$\frac{\partial^2 P}{\partial^2 S} = \frac{\partial^2 C}{\partial^2 S}$$

[1]

- iii. Theta is the rate of change of the value of the portfolio with respect to the passage of time with all else remaining same. It is obtained by differentiating the put call parity by t i.e. $\frac{\partial P}{\partial t}$

$$\frac{\partial P}{\partial t} = rKe^{-r(T-t)} + \frac{\partial C}{\partial t}$$

[1]

- iv. The values and equations for the calculations are given in the table below:

The question only hints that the option is at the money, i.e. the current stock price and strike price are the same. To calculate delta stock price is not required. The successive sub - parts of the question expects the candidates to determine the value of the gamma and theta using a specific formula, and does not ask to calculate. The candidate can either use the expressions or might assume any value of current stock price to determine gamma, theta and interpret theta by directly assuming some value of current stock price.

2 –For calculating parameters and delta, 1- For identifying the relationship between delta, gamma, theta and value of the option, 3 -Computing gamma, value of the option and theta from above expression, 1 – For interpretation of theta

The solution assumes stock price of Rs. 300:

S=K (at the money)	300
R	6%
Volatility	25%
T	0.5
d1	0.2580940
d2	0.0813173
Delta call	$N(d1) = 0.6018$
Delta Put	$N(d1) - 1 = -0.3982$
Price of a call option c	$S_0N(d1) - K\exp(-rt)N(d2)$
d1	$[\ln(S/K) + (r + 1/2\sigma^2)T] / \sigma T^{1/2}$
d2	$[\ln(S/K) + (r - 1/2\sigma^2)T] / \sigma T^{1/2}$
N(d1)	0.6018328
N(d2)	0.5324052
Price of a call option	25.5487730
Price of a put option	16.6824330
Put call parity	$P + S = K \exp(-rt) + C$
Gamma of both put and call	$\phi(d1) / S\sigma T^{1/2} = 0.00728$
Relationship between theta, gamma and delta given V = value of portfolio	$\Theta + rs\delta + 1/2 \sigma^2 S^2 \gamma = rV$
Θ of call calculated from the above equation	-29.76411525 = -0.08155 per calendar day
Θ of put calculated from the above equation	-12.29609565 = -0.03369 per calendar day
Theta is usually negative for an option. This is because as the time to maturity decreases with all else remaining the same, the option tends to become less valuable	

[7]

[10 Marks]

Solution 6:

i.

	Risk neutral	Real world
Default probability p	0.01	0.30%
Face value	100	100
T	1	1
Expected payoff	99	99.7
rate of return	6%	R
Value of a bond	93.23468882	$99.7 * \exp(-R * 1)$
Value of a bond	Value of a bond is the expected payoff in a risk neutral world discounted at the risk free rate	Alternatively the value of the bond in the real world is the expected payoff at some different risk adjusted rate

For the two prices to be same, we have $93.23468882 = 99.7 * \exp(-R)$,

Which implies $\exp(-R) = 0.935152345$

Or $-R = -0.067045827$ Or $R = 0.067045827$

Which implies $R - r = 0.705\%$, $R_M - r = 4\%$, from security market line,

Beta = $0.705\% / 4\% = 17.6\%$

[4]

ii.

Risk free rate	Coupon	Bond yield	Recovery probability
6%	7%	6.60%	30%

Risk free value of the bond = $3.5 / (1.03) + 103.5 * 1 / 1.03^2 = 100.9567348$

Actual bond price = $3.5 / (1.033) + 103.5 * 1 / 1.033^2 = 100.3810366$

Let q be the probability of default.

The present value of cost of defaults equals excess of the price of the risk free bond over the price of the corporate bond with same maturity and paying the same coupon.

The expected value from default = $(1-q) * 1.035 / 1.03^2 + 3.5 / 1.03 + q * 0.30 * 103.5 / 1.03^2$

Which implies the price of the bond should be such that it allows for the default risk.

This implies:

$100.381037 = (1-q) * 1.035 / 1.03^2 + 3.5 / 1.03 + q * 0.30 * 103.5 / 1.03^2$

Or $q = (100.9567348 - 100.3810366) / ((1-R) * 97.55868) = 0.00843$

[4]

[8 Marks]

Solution 7:

i. AR(1) – Auto regressive process of order 1 with parameter α [1]

ii. Define $Y_n = X_n / \alpha^n$

By given condition:

$$\alpha^n Y_n = \alpha \alpha^{n-1} Y_{n-1} + e_n$$

$$\Rightarrow Y_n = Y_{n-1} + e_n / \alpha^n$$

$$= Y_{n-2} + e_n / \alpha^n + e_{n-1} / \alpha^{n-1}$$

$$= \dots$$

$$= Y_0 + \sum_{l=1}^n e_l / \alpha^l, l = 1 \text{ to } n.$$

So, $E(Y_n) = Y_0$, since $E(e_i) = 0$ for all i .

$$\text{Thus } E(X_n) = \alpha^n E(Y_n) = \alpha^n Y_0 = \alpha^n X_0 / \alpha^0 = \alpha^n X_0$$

$\text{Var}(Y_n) = \sum \text{Var}(e_i) / \alpha^{2i} = \sigma_e^2 \sum (1 / \alpha^{2i}) = \sigma_e^2 (\alpha^{2n} - 1) / [\alpha^{2n} (\alpha^2 - 1)]$, using sum formula for geometric series.

$$\text{Alternatively, } \sigma_e^2 \sum (1 / \alpha^{2i}) = \sigma_e^2 (1 / \alpha^2 + 1 / \alpha^4 + 1 / \alpha^6 + \dots 1 / \alpha^{2n}) = \sigma_e^2 1 / \alpha^2 (1 + 1 / \alpha^2 + 1 / \alpha^4 + \dots 1 / \alpha^{2n}) = \sigma_e^2 AS \text{ where } A = 1 / \alpha^2 \text{ and } (1 + 1 / \alpha^2 + 1 / \alpha^4 + \dots 1 / \alpha^{2n}) = S$$

Now $S = (1 + 1 / \alpha^2 + 1 / \alpha^4 + \dots 1 / \alpha^{2n})$ or $1 + AS = 1 + a + a^2 + \dots a^n$ or $S = (1 - A^n) / (1 - A)$.

$$\sum \text{Var}(e_i) / \alpha^{2i} = \sigma_e^2 AS = \sigma_e^2 AS = \sigma_e^2 1 / \alpha^2 (1 - 1 / \alpha^{2n}) / (1 - 1 / \alpha^2)$$

$$\text{Hence } \text{Var}(X_n) = \alpha^{2n} \text{Var}(Y_n) = \sigma_e^2 (\alpha^{2n} - 1) / (\alpha^2 - 1)$$

[5]

iii. An Ornstein Uhlenbeck process is given by

$$X_t = X_0 e^{-\gamma t} + \sigma \int_0^t e^{-\gamma(t-s)} dB_s \text{ with drift } \gamma x \text{ and diffusion } \sigma^2$$

$$E(X_t) = X_0 e^{-\gamma t}$$

Since $B_t - B_s$ follows $N(0, t-s)$

$$V(X_t) = \sigma^2 \int_0^t e^{-2\gamma(t-s)} ds \text{) using problem 4.}$$

$$\text{Let } y = -2\gamma(t-s)$$

$$\text{Then variance} = \sigma^2 \frac{1}{2\gamma} \int_{-2\gamma t}^0 e^y dy$$

$$= \sigma^2 \frac{1}{2\gamma} [1 - e^{-2\gamma t}]$$

If $\alpha = e^{-\gamma}$, $\sigma_e^2 / (1 - \alpha^2) = \sigma^2 / 2\gamma$ then the first two moments of AR(1) and Ornstein Uhlenbeck process coincide.

For AR (1), $E(X_n) = \alpha^n X_0 = X_0 e^{-\gamma n}$ which is the mean of Ornstein Uhlenbeck process

$$\text{Var}(X_n) = \sigma_e^2 (1 - \alpha^{2n}) / (1 - \alpha^2) = \sigma^2 \frac{1}{2\gamma} (1 - e^{-2\gamma n})$$

[4]

- iv. X_t displays mean reversion tendency as the drift coefficient is $-\gamma x$. Since γ is positive, the drift acts in opposite direction to the current value of X_t . This causes X_t to revert to zero.

[1]

[11 Marks]

Solution 8:

- i. 100% in A – expected return of 12%

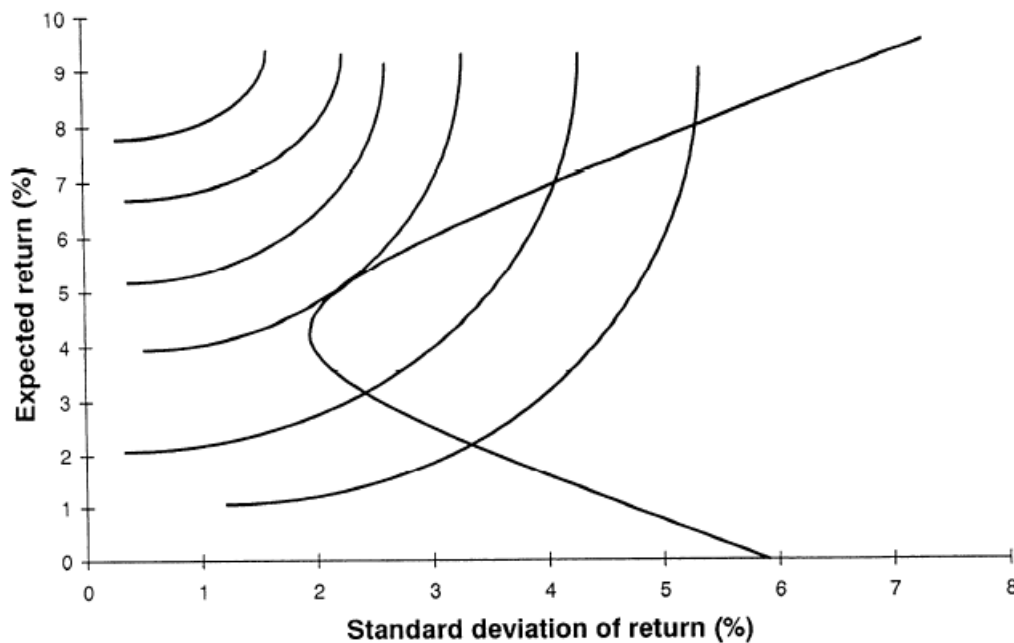
[1]

- ii. Proportion in A = $(V(A) - \text{Cov}(A,B))/(V(A) + V(B) - 2\text{Cov}(A,B))$
 $= (0.15 - 0.5 \times 0.30 \times 0.15)/(0.30 + 0.15 - 2 \times 0.5 \times 0.30 \times 0.15)$
 $= 18.47\%$

Proportion in B = 81.53%

[3]

- iii. Plot indifference curves in return-standard deviation space. Utility is maximised by choosing the portfolio on the efficient frontier where the frontier is at a tangent to the indifference curve.



[3]

[7 Marks]

Solution 9:

- i. The variance of the portfolio can be written as :

$$V = \text{var}[R_p] = \sum_i \sum_j x_i x_j C_{ij}$$

This can be expressed as -

$$V = \sum_i x_i^2 V_i + \sum_j \sum_{i \neq j} x_i x_j C_{ij}$$

Where all assets are independent, the covariance between them is zero and the formula for variance

becomes:

$$V = \sum_i x_i^2 V_i$$

If we assume that equal amounts are invested in each asset, then with N assets the proportion invested in each is $1/N$. Thus:

$$\sum_i (1/N)^2 V_i = (1/N) \sum_i (1/N) V_i = \frac{V^*}{N}$$

where V^* represents the average variance of the stocks in the portfolio. As N gets larger and larger, the variance of the portfolio approaches zero. This is a general result – if we have enough *independent* assets, the variance of a portfolio of these assets approaches zero.

[3]

- ii. In real world it's a difficult condition to meet in most markets since the correlation coefficient and the covariance between assets is positive.

[1]

- iii. With equal investment, the proportion invested in any one asset x_i is $1/N$ and the formula for the variance of the portfolio becomes:

$$V = \sum_i (1/N)^2 V_i + \sum_j \sum_{i \neq j} (1/N)(1/N) C_{ij}$$

Factoring out $1/N$ from the first summation and $(N-1)/N$ from the second yields:

$$V = (1/N) \sum_i (1/N) V_i + \frac{(N-1)}{N} \sum_j \sum_{i \neq j} \frac{C_{ij}}{N(N-1)}$$

Replacing the variances and covariances in the summation by their averages V^* and C^* , we have:

$$V = \frac{V^*}{N} + \frac{N-1}{N} C^*$$

The contribution to the portfolio variance of the variances of the individual securities goes to zero as N gets very large. However, the contribution of the covariance terms approaches the average covariance as N gets large.

So, as the number of assets in the portfolio is increased, the variance of the return on the portfolio gets closer to the average covariance of return between the pairs of assets in that portfolio.

[3]

[7 Marks]

Solution 10:

- i. The field of behavioural finance looks at how a variety of mental biases and decision making errors affect financial decisions. It relates to the psychology that underlies and drives financial decision-making behaviour.

Behavioural finance challenges the assumption of expected utility theory that decisions are made on the basis of maximising the expected value of utility under the investor's particular beliefs about the probability of different outcomes.

[2]

ii.

- a) Anchoring – although the data shows that capital appreciation is unlikely to continue, the investor is anchored that history will repeat itself and therefore experts opinion is not given due consideration.
Overconfidence – past evidence of the investor's ability to make money means could have become overconfident in his own abilities.
- b) Hindsight bias – Events that happen will be thought of as having been predictable prior to the event, i.e. re-evaluating past evidence to prove the scenario was always likely to occur.

[5]

[7 Marks]

Solution 11:**i. Desirable characteristics of Term- structure model**

- a) The model should be arbitrage free
- b) Interest rates should be positive.
- c) $r(t)$ and other interest rates should exhibit some form of mean-reverting behaviour.
- d) Should be able to calculate the prices of bonds and certain derivative contracts easily.
- e) The model should produce realistic dynamics
- f) Should fit the historical interest rates data with appropriate parameter estimates.
- g) Should calibrate easily to current market data.
- h) Should be flexible enough to cope properly with a range of derivative contracts

[4]

- ii. If the short-rate changes, the volatility of the process is unchanged in the Vasicek model. In the CIR model (an increase in the short rate will lead to an increase in the volatility).

[3]

- iii. The equations defining the two models are similar:

Vasicek: $dr(t) = \alpha[\mu - r(t)]dt + \sigma d\tilde{W}(t)$

H&W: $dr(t) = \alpha[\mu(t) - r(t)]dt + \sigma d\tilde{W}(t)$

Common Features:

- Both are continuous-time Markov models of the short rate of interest
- Both are Ito processes defined by a stochastic differential equation
- Both the models are one-factor models (ie they incorporate only one source of randomness)
- Both models are usually defined in terms of a standard Brownian motion under the risk-neutral probability measure.
- Both models imply that the short rate is mean-reverting.
- Both models imply that the future short rate has a normal distribution.
- Both models allow negative values for the short rate.

Key Differences:

- The key difference is that the Vasicek model is time-homogeneous (with constant m), whereas the Hull & White model is not (with time-dependent m).
- The Hull & White model has to be calibrated (i.e. a function has to be selected for $m(t)$) to match the current pattern of bond prices.

- Both models are mathematically tractable, although the Hull & White model is slightly more complicated algebraically.

[5]

[12 Marks]

Solution 12:

- i. It is a statistical measure of downside risk.

[2]

- ii. Advantage: normal distribution is easy to manipulate to calculate VaRs based on only two parameters.

Disadvantage: results may be misleading with skewed or fat tailed distribution.

[2]

[4 Marks]
